Origami DNA
If you do not have a gene sequencer at home, just make DNA out of paper origami by following this tutorial! **https://dgit.in/aug20-19**

10 years of DNA origami
A video showing the process of DNA origami, as well as advances over 10 years in the field. **https://dgit.in/aug20-20**

mated and did not require any manual intervention. First, the algorithm breaks down the given shape into a series of polygons. Then, it creates a scaffold through the structure using a single DNA strand which provides the basic shape and holds the entire molecule together. The resulting particles are long lived and stable.

Researchers from Harvard have also created algorithms that allows DNA to be formed into particular shapes. One of their approaches breaks down the target shape into small components, then builds them up using pieces of known DNA shapes, similar to the process of putting together a LEGO set.

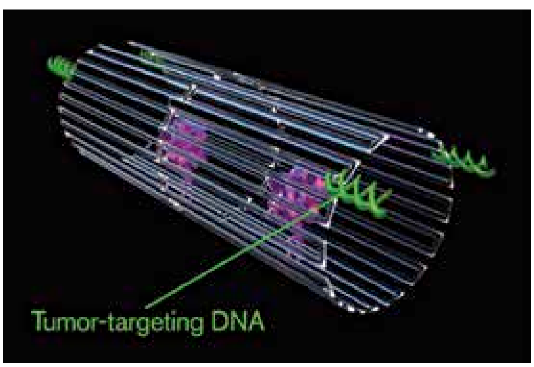
NANOMEDICINE

Within the immune system, there are antibodies that flag problems such as viruses or cancerous cells, and call in other cells to tackle or eat up these problematic cells. Monoclonal antibody are artificially produced antibodies that perform the same task as natural antibodies. Therapy using monoclonal antibodies can be used to treat cancer, for example, as they boost the immune response. However, monoclonal antibodies are poorly soluble in water, cannot easily pass cellular barriers, have a short life, are toxic, and degrade rapidly once introduced. Using them in treatments can be quite challenging but DNA origami promises to change that.

The structures made from DNA can deliver the required dose of monoclonal antibodies reliably and safely to target locations within the body. Researchers from Harvard were able to coat the DNA origami with chemicals that allowed them to maintain their structure within the body and not dissolve into individual strands within the cells. Then, by testing a variety of different shapes, they were able to identify the structures that were most suitable for delivery of target drugs.

Credit: Wyss Institute at Harvard University

Three dimensional shapes created using DNA. Each cylinder represents a double helix.

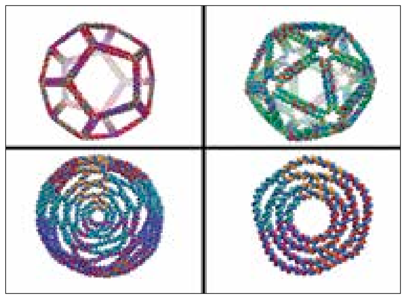


Credit: ASU

The tumour targeting DNA nanorobots

Over a course of five years, researchers from Arizona State University and the Chinese Academy of Sciences developed an entire programmable robotic system that uses DNA origami to fight tumors, by cutting off the blood supply that allows them to grow. The whole system is surprisingly simple. First, blood clotting molecules, called thrombins, are attached to a DNA scaffold. This scaffold folds up into a cylinder to travel through the bloodstream. At the edges of this cylinder are a series of small sensors called DNA aptamers. These DNA aptamers target a specific protein called nucleolin, which is only found on the surface of tumour cells. When the protein is detected, the DNA aptamers attach to the inside of the blood stream, unfolding the scaffold, and exposing the thrombin to the blood. The blood then clots up, preventing supply to the tumour. In this way, the cancerous cells get destroyed but the healthy cells are not affected. When tested on mice and pigs, the nanorobots could seriously damage tumours within 24 hours without any harmful side effects. Three out of eight mice tested with the therapy showed complete regression of all tumours.

At MIT, the same team that has been making 3D shapes have also been testing a variety of DNA particles for provoking an immune response from the body. Contrary to what is expected, the strongest immune response is not provoked by simply packing a DNA scaffold with as many antigens as possible. Instead, the position and orientation of the antigens within the DNA structure has to be finely tuned, which takes a lot of trial and error. The team has been replicating the immune response to viruses by structurally mimicking known viruses using DNA origami and antigens. While the research had been focused on developing therapies for HIV, there is now active research towards finding a vaccine for SARS-CoV-2. For this purpose, the researchers swapped out the antigen derived from HIV and began using one that was sourced from novel coronavirus. The cells are now being tested on mice to check the immune response. As the antigens from any other coronavirus mutations can also be attached to the same underlying DNA scaffold, it is possible to use the same approach to target many variants of coronavirus, including ones that might emerge in the future.



Credit: MIT

Complex 3D structures created by researchers at MIT